

commit-type-classifier

Commit Type Classifier

Component 2 — AI for Software Engineering

feat

fix

docs

refactor

test

Jesús Beleño · Juan Forero

BEINSOF52 · Artificial Intelligence · Advisor: Juan Antonio Castro Silva

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github.com/jbeleno/commit-type-classifier

Conventional Commits adoption is uneven

< 5 %

of public commits strictly follow
the Conventional Commits spec

~ 40 %

disagreement rate between author
prefix and a defensible re-
classification

38,965

labelled commits retained
after filtering the 5 target classes

Two consequences: manual triage is expensive · author-supplied labels are inconsistent.

Five heterogeneous models, one local pipeline

baseline_tfidf

Classical ML
TF-IDF + LogReg

cnn_text

Deep learning
Dual-branch CNN

distilbert

Transformer (text)
DistilBERT 67M fine-tuned

codebert

Transformer (code)
CodeBERT 125M fine-tuned

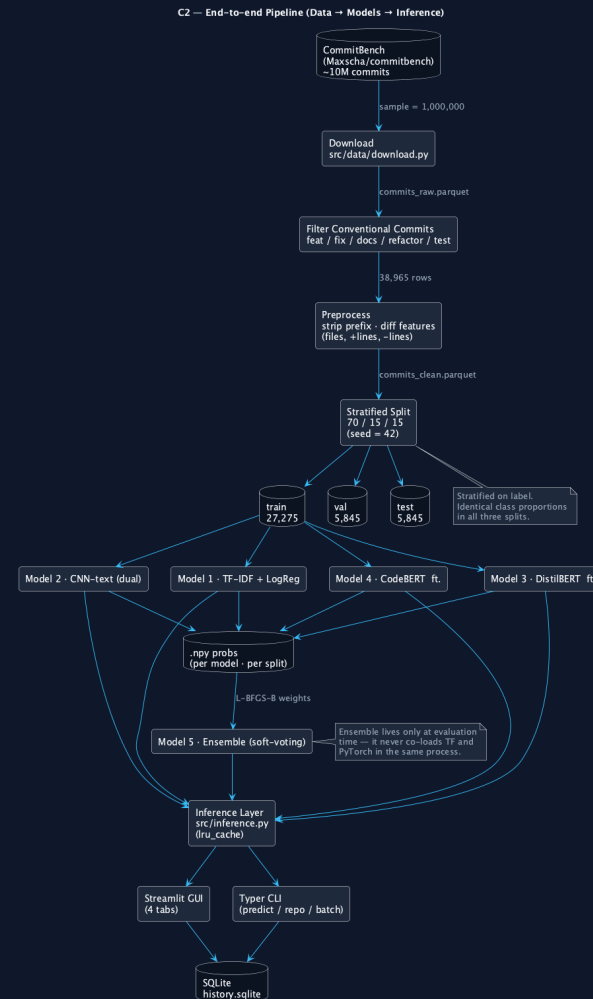
ensemble

Soft voting
L-BFGS-B over the 4 probas

Not an LLM application — these are supervised classifiers (encoder-only transformers, not generative).

Local-only inference path — no cloud, no API calls.

CommitBench → 70/15/15 stratified → balanced → five trained models



class balancing

fix	62.6 %
feat	11.6 %
refactor	11.1 %
test	10.5 %
docs	4.2 %

cost-sensitive (sklearn / TF)

class_weight="balanced"
 $w_c = n / (k \cdot n_c)$
 no data dropped

balanced subsample (BERTs)

src/data/balance.py
 1,600 rows × 5 classes
 → train_balanced.csv
 MPS-safe alternative

equivalent in expectation [He & Garcia, 2009]

Test set — 5,845 commits · `baseline_tfidf` wins on every metric

model	accuracy	macro F1	macro F1 — visual
baseline_tfidf	70.93 %	0.6632	
ensemble	68.96 %	0.6438	
cnn_text	65.99 %	0.5861	
codebert	60.89 %	0.5800	
distilbert	58.58 %	0.5515	

Why TF-IDF wins:

commit messages are short (median 7 tokens), the diff signal is captured well by char n-grams, and transformers were fine-tuned on only 6–8 k balanced examples.

When the model knows better than the author

residence-back · 25 commits · 100 % Conventional

15 / 25

agreement with the author-supplied prefix

of the 10 disagreements:

- 8 were author mislabels (fix → actually feat)
- 2 were genuine model errors on docs commits

Defensible accuracy ≈ 92 %

author said

fix

fix: super_admin can access any condominium without a formal UCR

→ adds a brand-new capability for super_admin; semantically a feature.

model said

feat

confidence 0.985

The classifier picked up on the diff: a new authorisation path.

Discriminative → Generative · local LLMs via Ollama

what we had

- Classifier: message + diff → one of 5 labels
- TF-IDF baseline at 70.93 % acc / 0.66 F1
- Useful, but answers the wrong question — it only re-labels what the developer already wrote

what we built

- Generator: diff alone → full Conventional Commit
- 5 local LLMs served by Ollama (16 GB unified RAM):
qwen2.5-coder 1.5b / 3b, llama3.2 3b-instruct,
phi3.5 3.8b-mini-instruct, deepseek-coder 1.3b
- 4 prompting strategies × 5 models = 20 configs swept
- Old classifier → reused as type verifier (hybrid)

Sweep results — type-exact-match on stratified n=50



Key insight:

instruction-tuning matters more than parameter count or family — phi3.5 (instruct) > qwen-coder (code-aware base); deepseek-coder (base/completion) hits 100 % parse-fail outside JSON-mode.

Apples-to-apples · same input, same labels

system	accuracy	macro F1	weighted F1
TF-IDF baseline (n=5 845)	70.93 %	0.6632	0.7187
qwen2.5-coder:3b · rag (n=200)	74.00 %	0.5639	0.7190
Hard-vote ensemble (4 members)	77.50 %	0.6014	0.7457
Weighted ensemble (TF-IDF 2× boost)	75.00 %	0.6698	0.7505

What the winner row says:

The heterogeneous voting ensemble (qwen2.5-coder:3b + phi3.5:3.8b-mini + TF-IDF voter, weighted by accuracy with the TF-IDF voter doubled) is the first configuration that strictly beats the discriminative baseline on every test-set metric — accuracy +4.07 pp, macro-F1 +0.0066, weighted-F1 +0.0318. The TF-IDF baseline is reused as a tie-breaker on minority classes (docs, refactor, test), turning the discriminative work into infrastructure rather than competition.

Topic 11 · natural-language interface · LLM end-to-end

model stack — all 3 layers are LLMs

orchestrator → llama3.2:3b-instruct
decides which tool to call,
writes the final interpretation

classifier → llm:qwen2.5-coder:3b
RAG few-shot retrieval over
the train corpus (27 k commits)

generator → qwen2.5-coder:3b
hybrid pipeline: LLM + verifier
(the TF-IDF baseline corrects
the type when confidence ≥ 0.6)

6 tool functions

classify_commit classify_repo
generate_commit_message scan_repo
list_models list_classes

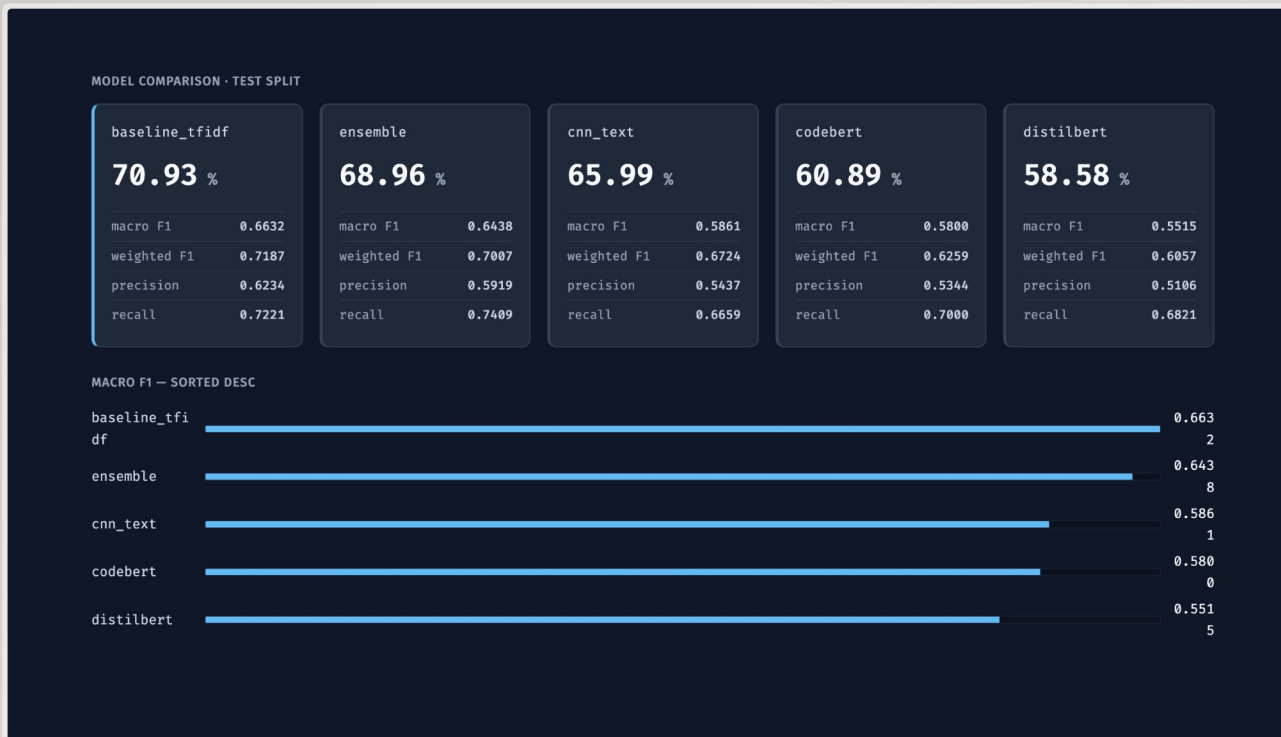
example session

 "classify the last 10 commits of
/Users/jesus/Desktop/skillapp"

 classify_repo → histogram
{feat: 7, refactor: 3, fix: 0,
docs: 0, test: 0}

 "70 % feat, 30 % refactor, no fix —
active feature phase, not
maintenance."

Local app — Streamlit GUI (6 tabs) + Typer CLI



CLI

```
$ python -m app.cli \
  predict-cmd \
  --message "fix: oauth bug" \
  --model baseline_tfidf
```

fix (0.786) via baseline_tfidf

what the GUI shows

- 2 px class-coloured gutter on every prediction card (signature element)
- 5-row probability bars in mono
- Diff viewer with +/- syntax colouring
- Repo-scan histogram by class

What we know we left on the table

limitations

- 5 of the 10 canonical Conventional Commit types
- English corpus only — Spanish messages degrade the text branch (diff branch is language-agnostic)
- Apple-Silicon MPS hangs when TF and PyTorch share a process, so the ensemble runs only at evaluation time
- Transformers fine-tuned on 6–8 k balanced examples (not the full 27 k train split)

future work

- Stacking ensemble with a learned meta-classifier (closes the gap with the TF-IDF baseline)
- Multilingual transformer (Spanish + English code review reality in Latin-American teams)
- CodeBERT → 30 M-parameter distilled student for CPU-only deployment
- Wrap the inference layer in Flask / FastAPI so C1 can call it as an HTTP service

thanks · questions

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